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COMMONWEALTH OF AUSTRALIA

PATENT SPECIFICATION

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Complete Specification Entitled PROCESS AND COMPOSITIONS FOR PROTECTING MATERIALS AGAINST FUNGAL AND BACTERIAL INFECTIONS AND PROTECTED MATERIALS OBTAINED THEREBY.

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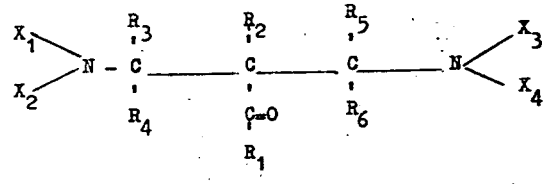
Related Art: Nil.

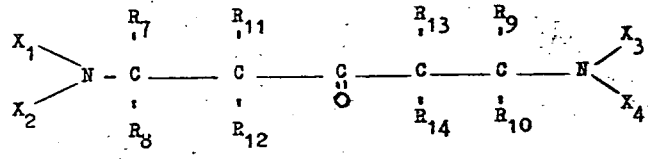
The following statement is a full description of this invention, including the best method of performing it known to us :

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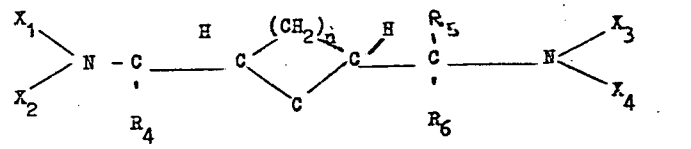
The present invention relates to a process for protecting materials of various kinds against infection and damage by micro-organisms, as by bacteria and funti, and to antimicrobial compositions ror use in such process.

We have discovered that compounds obtained by condensation of secondary amines with ketones and aldehydes of the following formulas:

I. 

II. 

and

III  ,

and their salts, wherein:

R_1 to R_{10} are each a member of the group consisting of hydrogen, alkyl having up to 18 carbons, alkenyl having up to 18 carbons, aralkyl and cycloalkyl having up to 9 carbons, except that at least one of R_{11} to R_{14} is not hydrogen; while R_2 can also be lower hydroxyalkyl;

X_1 to X_4 are each a member of the group consisting of hydrogen, alkyl having up to 18 carbons, alkenyl having up to 18 carbons, hydroxyalkyl, cycloalkyl having up to 9 carbons, mononuclear aryl and aralkyl having up to 9 carbons; while at least one of the pairs of X_1 and X_2, and X_3 and X_4, may be part of a saturated heterocyclic ring containing the indicated N together with carbon also a further hetero atom of the group consisting of oxygen, nitrogen and sulfur; and n

is an integer from 1 to 6 inclusive, possess unexpected and highly useful antimicrobial activity and can accordingly be employed to protect industrial products and intermediate preparations of wide variety against the deteriorating action of bacteria and fungi.

In the above formulas, the alkyl and alkenyl groups are preferably lower alkyls and lower alkenyls having up to 6 carbon atoms, but they may have up to 18 carbons; while the cycloalkyl groups include cyclopentyl and cyclohexyl and up to cyclononyl; and aralkyl includes thephenyl- and naphthyl substituted lower alkanes wherein the phenyl or naphthyl group may or may not be substituted by lower alkyl, chloro, bromo, hydroxyl and other groups.

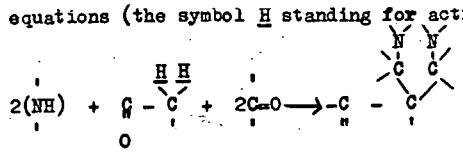
Many of the compounds embraced by the above formulas are known as such, but so far as we are aware they have not been known to possess unusually high antimicrobial activity and have not been made available in the form of compositions prepared for use for the disinfection of various materials subject to fungal and bacterial attack.

One of the outstanding advantages of the compositions of the present invention is that while they are characterized by high potency, they are free from toxic metals, like mercury and copper, which are injurious to human beings and the lower animals. A highly effective antimicrobial composition for industrial use which was free from toxic metals, particularly mercury, has long been sought but the non-metallic compounds and compositions heretofore suggested have not been comparable in potency to the known organic mercurials. The compositions of the present invention, on the other hand, in addition to being free of metal, have an activity of the order of that of many mercurial compounds and even excel many mercurial and other metal compounds now in use.

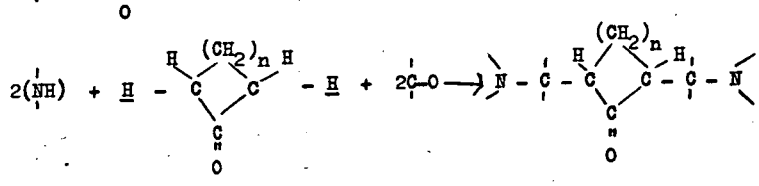
The compounds can be synthesized by techniques involving a ketonic compound which must contain at least two active hydrogen atoms on a single carbon atom joined to the carbonyl group, or on two cyclic carbon atoms joined to the common carbonyl, by which active hydrogens the ketonic compound

(sometimes hereinafter referred to as the "parent" ketone) is condensed with a secondary amine and with an aldehyde or another ketone, as indicated by the following equations (the symbol H standing for activated hydrogen):

IV.

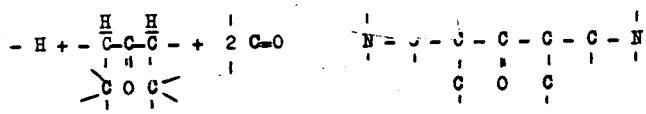


V.



(wherein n is as above defined), or on two carbon atoms which contain at least one active hydrogen atom each and are joined to the common carbonyl group, as represented in the following equation:

VI.



The introduction of two methano nitrogen groups in the condensation products may take place symmetrically as shown in equations V and VI to yield one such group on each side of the carbonyl, or asymmetrically with

both methano nitrogen groups on a single carbon atom connected to the carbonyl as shown in equation IV.

However, the symmetrical condensation products obtained by condensation of the simplest ketone, acetone, with formaldehyde and a secondary amine were found practically of no utility as antimicrobials and are specifically excluded. On the other hand, the corresponding non-symmetrical compounds obtained with acetone as the "parent" ketone were found to exhibit an extremely high activity.

Surprisingly, the symmetrical products obtained, as in equations V and VI which are chemically similar to those produced in equation IV, show unusual and unexpected utility when incorporated in antimicrobial compositions and tested under strenuous conditions both in the laboratory and on an industrial scale.

The compounds are prepared by reacting substantially molar ratios of 2:1:2 with respect to secondary amine, active hydrogen-containing ketone, and aldehyde or second ketone; however, a considerable excess of the first and/or third reaction component may be used.

The syntheses of these products can be conducted in a variety of solvent systems including water, alcohols, ethers, inorganic and organic acids, as well as in the excess of one of the reagents, e.g., of the amines and aldehydes or ketones. The introduction of the two nitrogen-containing groups can be accomplished sequentially if the two groups are different, or in a single step for identical groups. The net reaction involves at least two moles of secondary amine, two moles of aldehyde or ketone, and one mole of parent or active hydrogen-containing ketone. The amines can be used in the reaction in the form of salts such as the acetate or hydrochloride, or as free bases. Reaction conditions are chosen to provide for the introduction of the two basic groups, as is well known for conducting the "Mannich condensation". This requires, in general, elevated temperatures, reaction periods from 1 to 24 hours, the use of solvents to provide reagent contact and excesses of the amine and aldehyde or ketone over the exact molar ratios described and compared to the parent ketone. Isolation of products may be accomplished by the usual techniques of crystallization or distillation. Other chemical groupings may be present in the reaction components, such as alkyl and halogen, provided they do not contain active hydrogens which will interfere in the condensation reaction. Usually such active hydrogen atoms

can be first blocked by such masking techniques as alkylation, ether or ester formation, and the like, as is well known to chemists.

For the application of the condensation products to the preservation or mildew proofing of such varied systems as adhesives, paints, cutting oils, water-flooding for secondary oil recovery, slime control in paper mill water systems, wood treatment formulations, agricultural compositions for control of micro-organisms, water and air conditioning units, etc., the products can be added directly, in water or solvent systems, or in the form of salts prepared from the bases and inorganic or organic acids, phenols, substituted phenols and other salt-forming compounds containing acidic hydrogens. Also, the nitrogen containing groupings can be quaternized with such agents as lower alkyl halides, such as methyl, ethyl, propyl and butyl chlorides, bromides and iodides, dialkyl sulfates like dimethyl, diethyl, dibutyl and dilauryl sulfates, alkyl p-toluene sulfonates, such as methyl, ethyl, propyl, butyl, amyl and hexyl p-toluene sulfonates, long chain halides such as lauryl, myristyl and stearyl chlorides, bromides and iodides, particularly lauryl bromide and dodecyl chlorides, aralkyl halides, like benzyl and phenyl propyl and other phenyl-lower alkyl chlorides, bromides and iodides, etc. Water or oil-soluble or dispersible products are thereby obtained.

The activity of these products against bacteria and fungi has been demonstrated over a wide range of pH, namely 3 to 12, both in laboratory tests and in field trials. The compositions need not be prepared from pure compounds; the presence of other reaction by-products which have practically no biological action do not adversely affect the active agent.

The mode of action of these products cannot be attributed to a simple mechanism. It has been frequently reported that condensation-type products can slowly yield under hydrolytic conditions the original reactants and these supply the basis of properties either physically or chemically. The compositions of this invention show, e.g., such high orders of antimicrobial activity that none of the original reaction components, not even the formaldehyde, which is the preferred linking agent between the amine and the "Parent" ketone, could be the responsible factors.

A hypothesis may be that these products become degraded to yield unsaturated ketones, compounds known to show antimicrobial activity under certain conditions. However, the incorporation of our compositions in such acid systems as certain of the latex paints, paper mill water, or even in biological laboratory tests conducted on the acid side, which would prevent such degradation, nevertheless gives results of excellent order of control of growth of microorganisms. It is well established that these bases remain intact as salts under such acid conditions, and tests for the presence of alpha, beta-unsaturated ketones have proved to be negative.

As indicated above, the amines employed in preparing the condensates can be cyclic amines in which each N of the formulas is part of a ring. As disclosed hereinafter, the amine groups in each molecule of a condensation product can be of different types; thus, one group can be cyclic and the other not, or they can be different cyclic groups as well as different non-cyclic groups. The cyclic amines include morpholine, piperidine, piperazine, pyrrolidine, parathiazine, meta-thiazolidine and other, preferably saturated, cyclic amines with or without an additional hetero atom.

One of the most useful applications which we have discovered for the compounds employed in the present invention is in the preservation and disinfection of emulsions formed from "soluble oils". "Soluble oils" are industrial compositions used in metal-working as coolants, lubricants, flushing agents, and anti-corrosives, after being mixed with and emulsified in water. Other applications are as fluids in hydraulic systems and lubricants in extrusion processes. Where metals contact metals, as in the case of cutting tools in lathes, grinders, and drill presses, or even where non-metals are extruded through metal dies, the use of these emulsified oils is essential.

The oils consist generally of chemically complex mixtures of petroleum, emulsifying agent, and coupling agent. (L.R. Sabina and H. Pivnick, Applied Microbiology, Vol. 4, p. 173 (1956)). The petroleum fraction is composed of aliphatic or cyclic compounds of varying molecular weight and degree of complexity. Fats used may be lard, sperm oil, rape seed oil, etc. Both petroleum and fat may be chlorinated, sulfurized or sulfonated. Emulsifying agents may be amine or sodium soaps of high molecular weight, alcohols, poly-

ethoxylated phenols, etc. Coupling agents are used in "soluble" oil emulsions to improve the stability of the emulsions. Alcohols, glycols, and glycol ethers are typical.

It has been observed that these emulsions are capable of supporting the growth of a tremendous variety of micro-organisms, both pathogenic and non-pathogenic, aerobic and non-aerobic, and in sufficient population density under field use conditions to cause heavy losses in the need for frequent oil replacement as well as lost man-hours in equipment clean-ups and sickness due to dermatitis. Emulsions under microbial attack develop odor, become corrosive to metals, lose lubricating properties, and separate into phases. They must then be discarded and replaced with fresh emulsion long before their potential useful life has been realized. The "soluble" oils (emulsions) are of the type disclosed in the patents to Bennett, Nos. 2,976,244 and 2,976,236, both dated March 21, 1961.

A wide variety of chemicals have been screened for application as preservatives. Failure has been noted even with levels as high as 1,000 parts per million of emulsion with alkyl-dimethyl benzyl ammonium chlorides, difatty alkyl quaternary ammonium chlorides, tris-(hydroxymethyl) aminomethane, tetra-decylamine-ortho-phenylphenate, 2,6,8-trimethylnonanone-4, oleyl sapamine base, tetradecylamine, alkyl quaternary ammonium chlorides, alkylamine-ortho-phenyl phenate, 2-amino-1,4-naphthoquinone, decane aldehyde, dl-camphor, dialdehyde, dibenzylamine, diacbutyl ketone, 2-ethylhexaldehyde, ethyl-hexadecyl-2ketomorpholinium bromide and many other varied types. The relatively few products in use commercially have stringent limitations. Thus, one which contains sodium ethylmercury thiosalicylate as an active ingredient cannot be used in emulsion systems which contain reducing agents such as sulfur, sulfuretted oils, or sulfide producing organisms because of complete chemical inactivation. Another, which depends on the liberation of formaldehyde for its antimicrobial activity, is unstable in alkaline media, typical of soluble oil-derived emulsions and is stabilized by sodium bisulfite. The latter agent, however, lowers markedly the antibacterial activity of the compound requiring impractically high levels to be used.

In view of the types of chemicals which were failures, the limitations of those in use, and the highly varied types of organisms able to thrive in the emulsions under field conditions, it was completely unexpected that even one of the simplest members of the series of compounds disclosed above

should be capable of not only preserving emulsions but rendering them for all purposes sterile.

The method of laboratory evaluation was based on the work of H. O. Wheeler and E. O. Bennett (Applied Microbiology, Vol. 4, p. 125 (1956)). Commercial "soluble" oils were obtained free of inhibitors. These oils were representative of the major types in use, and the heavy duty ones contained appreciable amounts of sulphur or compounds containing sulphur in a reduced valence state. In 4-liter open mouth jars three liters of emulsion were prepared by mixing the soluble oils with tap water in a ratio of 1:20 or 1:40, depending on the manufacturers' recommendations. Incorporated as a preservative and disinfectant by either addition first to the soluble oil or the emulsion, 1,1-bis-(diethylaminomethyl)-acetone was added to give a concentration by weight of 1,000 ppm in the emulsion. Next, 3 gms of powdered iron were added, followed by 6 gms of spoiled emulsion, obtained from a machine in industrial use, and which contained an extremely varied and high microorganism population. Controls were set up in which other commercial preservatives were added according to the recommendations of the manufacturer as well as the same emulsions with no inhibitors added. Air was bubbled through the emulsions in a vigorous stream for 8 hours, followed by the stopping of the air supply and stagnation for 16 hours. The cycle was repeated each 24 hours; sufficient tap water was added as required to maintain the volume at the original 3 liters, replacing evaporation losses. Every 3 or 4 days each emulsion was sampled, appropriate dilutions were made for plating out, using tryptone glucose extract agar. Bacterial colonies were counted after 48 hours incubation at 37°C and recorded. At weekly intervals each emulsion was re-inoculated with 0.2% of spoiled emulsion obtained from an industrial machine. No further additions of inhibitor were made with our compound; the tap water additions made to the other commercial test emulsions were fortified with inhibitors as prescribed by the manufacturer. The tests were run until there was complete destruction of the emulsion, heavy odor development, and bacterial population density greater than 1,000,000 - 10,000,000 organisms per ml.

None of the other inhibitors tested gave protection to the emulsions greater than two months. Failure was observed in two days in the sulfur-containing oils for the mercury-type inhibitor. The stagnation cycle allowed the growth of sulfate-reducing organisms evidenced by the strong hy-

drogen sulfide odor as well as the grayish black iron sulfide color in the emulsions which contained sulfonated compounds. In these, the mercury compounds failed in less than a week. At 1:1,000 concentrations, the substituted phenols failed, as well as a host of other preservative-type chemicals. The emulsions which contained our basic ketone survived for more than one year, sterility being maintained within 24 hours after each weekly inoculum of spoiled emulsion. These results were so unusual that the experiment was repeated in duplicate and 1,1-bis-(dimethylaminomethyl)-acetone was also included in a separate test. The results again showed complete protection for periods of a year with an initial concentration of 1:1,000 of inhibitor in the emulsion and no further additions.

The laboratory results, both unusual and unexpected, were then used as a basis for actual plant evaluation. A machine shop was selected in which machinery installation conditions were such that the life expectancy of cutting oil emulsions was not better than 3 weeks by actual experience. The emulsions were broken, heavy in odor, corrosive to metals, and high in bacterial count. Emulsion which had been in use but kept stagnant in sections of piping showed high sulfide content and showed, on culturing, high populations of sulfate-reducing bacteria. Without any clean-up other than the usual draining of reservoirs, pipes, and troughs, fresh emulsion was added to the individual machine systems, lathes and drill presses. 1,1-bis(dimethylaminomethyl)-acetone at a level of 1:1,000 by weight was used in one system and the rest of the machines were supplied with the same emulsion containing the inhibitors currently in use. Within 3 weeks these machines had to be cleaned of spoiled emulsion, the usual signs of spoilage being most obvious. The machine systems in which the emulsion treated with our product was being used were successfully maintained for more than 3 months, the only additions of our inhibitor being made in new emulsion at 1:1,000 added to replace that lost by mechanical means. The emulsions were in excellent condition throughout, free of odor, low in color and at very low bacterial population levels. This period of test included week-end shut-downs which permitted the growth of anaerobic type organisms in the emulsions ineffectually inhibited.

Another area of use of our improved process and compositions is in oil field water flooding operations wherein many problems are created by the growth of such microorganisms as iron bacteria, sulfate reducing bacteria,

capsule or slime forming bacteria, as well as algae and molds in the water employed, or the water used for oil recovery may be suitable chemically and physically yet give trouble biologically. The various types of organisms present can cause plugging of oil sands due to insoluble sulfide formation by reduction of sulphate, and oxidation of iron to form gelatinous slime masses of ferric hydroxide. In addition, the metabolic products are capable of accelerating corrosion of equipment such as tanks, pipelines, and pumps. A method of laboratory evaluation was used based on the work of R.C. Allred, T.A. Mills, and H.B. Fisher, (Producers Monthly, p. 31-2 (1954)). A special medium was prepared which gave vigorous growth factors for sulphate-reducing bacteria. Ferrous sulphate, ferrous ammonium sulphate, sodium thioglycollate, and l-cystine were some of the nutrients included. Air was excluded by filling the test containers completely and closing with screw caps. Sufficient inoculum from a five-day broth culture of sulfate-reducing bacteria was added to give the medium a count of 500,000 bacteria per ml. The compounds to be evaluated were added in water to yield varying concentrations. Growth of the sulfate-reducing bacteria in the presence of the compound under test was indicated by an intense blackening of the medium. The test was run for at least 30 days after the controls, i.e., those tests to which no compound had been added, turned black. The authors note that "experience has shown that before a compound can be considered for use as a bactericide in the field, it must be able to inhibit the growth of sulfate reducing bacteria for a period of 30 days."

To give better correlation between laboratory results and actual field performance the medium was made up using brine, actually drawn from the field, in place of distilled water. Also, the inoculum was prepared from field-isolated sulfate reducing microorganisms. A typical set of data obtained under these conditions was as follows:

Compound Tested - 2,2'-bis-(dimethylaminomethyl)-cyclohexanone.

	Concentration (by weight) in medium, parts per million		
	5	10	25
Duration of Inhibition	None	2 days	Still good at 30 days

Similar types of bacteria have been found to be responsible for the heavy hydrogen sulfide odor and black spots found in kraft pulp slush contained in stagnant areas of storage chests and other components of paper

manufacturing equipment. In particular, there was isolated Desulfovibrio desulfuricans. Under test conditions, in which sterile kraft pulp slush was inoculated with the above organisms, and air was excluded by avoiding agitation, in the bottoms of the test containers heavy bacterial growth, strong hydrogen sulfide evolution and intense darkening resulted. The addition of less than 4 parts per million of 2,2'-bis-(dimethylaminomethyl)-cyclopentanone, or the corresponding cyclohexanone as well as nearly similar concentrations of compounds of type I, III, and V completely suppressed the growth of the sulfate reducing type organisms evidenced by the complete absence of hydrogen sulfide and darkening of the pulp. In actual mill trials in draft pulp systems, equivalent excellent control of microbial growth was achieved at comparable concentration levels. Even wider spectra of anti-microbial activity were observed in mill trials taking in the complete system including groundwood, sulphite and sulphate pulps, bleached and unbleached in furnishes at pH's ranging from 4.2 to 9. The pH can, however, be as low as 3.2.

The control of micrcorganisms in the manufacture of pulp and paper is extremely important because certain fungi and bacteria form slimy gelatinous masses which plug paper, coat screens and cause serious problems at various stages in the processing. When these masses break loose and are carried into the final chest, the quality and strength of the paper are affected. The addition of 2,2' bis-(dimethylamino methyl) cyclohexanone to the white water system in mills has resulted in excellent control of slime conditions by a marked reduction in the microbial population in such mills.

This compound has worked well in newsprint mills using groundwood and sulfite pulps, kraft mills, specialty mills using mixed chemical pulps and board mills using waste paper and chemical pulps. Excellent results were obtained in mills operating at pH's ranging from 3.2-9.0 and at headbox temperatures of 80-140°F. Effective control has been obtained by addition of the compound at various places in the mill including pulp storage areas, machine chests, white water chests, broke chests and directly into the headbox. Several techniques of addition have proven to be successful: shock dosage once a day, and addition at various hourly periods such as once every 8 hours, once every 4 hours etc. and also continuous dosage over the entire 24 hour period. The actual dosage required for a specific application varied from 1-100 ppm depending on the method of addition, pH, temperature, type of furnish and other variables particular to each mill.

By way of example, a typical application was made in a mill using soda, sulfite and bleached kraft pulps, clay and various other sizing agents. The 2,2'-bis-(dimethylamino methyl) cyclohexanone was added to the hydropulper dump chest once every 2 hours at 11 ppm based on total furnish. The system was maintained free from slime and the control could be further demonstrated by a comparison of the microbial population after addition to that prior to treatment.

Sample	Microbial Population as Organisms/ml *
Prior to treatment	500,000
2 hrs. after addition	3,500
3 hrs after addition	3,500
4 hrs. after addition	4,500
5 hrs. after addition	1,500
6 hrs after addition	3,500
7 hrs. after addition	2,500

* Testing procedure carried out according to TAPPI standards T-228, T-449 and T-63.

Slime control was equally well effected in a newsprint mill when the 2, 2'-bis-(dimethyl amino methyl) cyclohexenone was added directly to the head-box at a 4.7 ppm dosage. The addition in this case was made over a 2 hour period once every 8 hours. The microbial populations throughout the white water system were kept under good control. The paper manufactured showed no slime spots and the paper machine system was free from slime.

The following examples of substances found by us to be effective anti-microbial agents, and their methods of preparation, are presented by way of illustration only and not as indicating the scope of the invention:

Example 1

1,1-Bis-(Diethylaminomethyl)-Acetone

This compound was prepared in the manner described in J.A.C.S. 65, 972 (1943) by adding a solution of 210 ml. diethylamine in 400 ml. water to 300 ml. acetone, to which was then added without cooling 170 ml. 37% aqueous formaldehyde solution. The reaction evolved heat, and after standing 16 hours, the 2-phase reaction mixture was made alkaline by the addition of dilute sodium hydroxide, the water phase saturated with sodium chloride and the upper oil phase separated. The brine layer was extracted with three 400 ml. portions of ether; the ether extracts were combined and joined with the oil.

The ether solution was dried over magnesium sulphate, and the solvent removed by distillation. The residual oil was fractionated by distillation under reduced pressure to yield the product, b.p. 103-111°/6 mm.

The reaction was run similarly with the substitution of equimolar quantities of other organic bases in place of diethylamine, namely, dimethylamine, di-n-propylamine, di-isobutylamine, di-2-ethylhexylamine, di-allylamine, morpholine, piperidine, and piperazine. The isolation of the oily reaction product was readily achieved by phase separation. Purification, if desired, could be achieved by fractional distillation or by fractional crystallization of the salts prepared in anhydrous alcohol or ether with inorganic acids. The free bases containing two nitrogen-bearing groups substituted on the one carbon atom of the acetone molecule were high boiling oils of very slight color showing low water solubility and being readily soluble in the common organic solvents. Water-saturated solutions showed pH values above 10.

Example 2

2,2-Bis-(Dimethylaminomethyl)-3-hydroxypropionaldehyde

320 g. dimethylamine hydrochloride, 320 gms. 37% aqueous formaldehyde and 600 ml. water were mixed and reacted with 176 g. acetaldehyde. There was slight evolution of heat, and the mixture was allowed to stand for 40 hours. The pH was adjusted to 6.8 and the oil extracted with three 800 ml. portions of chloroform. The chloroform extracts were combined, the solution was dried over anhydrous magnesium sulphate, and the solvents along with unreacted components removed by fractional distillation in vacuo.

Example 3

2,2'-Bis-(Di-n-propylaminomethyl)-Cyclohexanone

To a mixture of 49 gms. cyclohexanone and 65 ml. water there were added 100 gms. di-n-propylamine keeping the temperature at 24-26°C. The reactants were mixed for 15 minutes and then 82 gms. 37% aqueous formaldehyde were slowly added so that the temperature did not exceed 30°C. The mixture was then slowly heated on a steam bath for two and one-half hours. It was cooled to 80°C, 20 gms. of powdered sodium chloride were added and dissolved in the aqueous phase. The upper oil layer was separated, the water layer cooled and extracted with several portions of benzene. The combined benzene extracts and oil were clarified by filtration and freed of solvent and unreacted components by fractional distillation in vacuo.

The reaction was repeated using equimolar quantities of other secondary amines in place of the di-n-propylamine. These included dimethylamine (40% aq. solution), di-2-ethylhexylamine, diethanolamine, di-allylamine, morpholine, and others. The bases thus obtained were soluble in dilute acids, insoluble in alkali, and easily miscible with most of the common organic solvents and oils.

Example 4

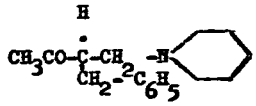
2,5-Bis-(dimethylaminomethyl)-Cyclopentanone Dihydrochloride

70 gms. Cyclopentanone, 55 gms. paraformaldehyde, 400 ml. glacial acetic acid, and 73 gms. dimethylamine hydrochloride were mixed and allowed to stand at room temperature for 16 hours. The mixture was then heated for 1 hour on the steam bath. It was clarified by filtration, and then mixed with a large volume of acetone and petroleum ether to precipitate the dihydrochloride of the 2,5-bis-(dimethylaminomethyl) cyclopentanone, m.p. 186-187°C after recrystallization from ethyl acetate-petroleum ether. Additional material could be recovered from the mother liquor.

Example 5

A mixed amine keto base can be prepared by stepwise reaction using two different secondary amines. Two condensation steps are thus effected which present variations in the aldehyde or ketone as well.

Step 1: N-Piperidino-2-benzylbutanone-3



This intermediate was prepared in the manner described in Archiv-Pharm. Vol. 264, p. 741 (1926) as follows:

33 gms. benzylacetoacetic ester were mixed with 380 ml. 2.5% potassium hydroxide solution until most of the ester dissolved. After periodic shaking and standing for 48 hrs., undissolved oil was separated and the mixture neutralized with hydrochloric acid. The well-cooled reaction mixture was then mixed with 18.3 gms. piperidine hydrochloride and 15 ml. 30% aqueous formaldehyde solution. After complete solution was attained, a total of 10 ml. conc. hydrochloric acid was added in small portions to cause vigorous evolution of carbon dioxide. A small amount of insoluble by-product was removed by ether extraction, the main solution made strongly alkaline and ether extracted. By fractional distillation at a b.p. 180-183°/14 mm, the product was obtained as the free base. Its hydrochloride prepared in ace-

tone had a m.p. of 145°.

Step 2: The N-piperidino-2-benzylbutanone-3 could be used as the free base or hydrochloride to react with a molar equivalent of a secondary amine and aldehyde or ketone to yield products having two different nitrogen-methane groupings joined to the same carbon atom. Examples include the reaction products obtained with the di-alkyl amines named in the previous examples and formaldehyde to yield N-piperidino-2-benzyl-2-dialkylaminomethyl-butanone-3 as well as other variations using aldehydes and ketones. The reaction conditions for Step 2 are the same as in Examples 1 to 4.

Example 6

Step 1: 2-(Dibenzylaminomethyl)-Cyclohexanone Hydrochloride.

This mono-substituted intermediate was prepared according to J. Org. Chem. Vol. 24, 1069 (1939). A mixture of 9.8 g. cyclohexanone, 1.5 g. of paraformaldehyde, 11.7 g. dibenzylamine hydrochloride, and 50 ml. acetic acid was heated for 2.5 hrs. at 55-65°C. The solvent was removed under reduced pressure at a temperature below 65°C. The oily residue was dissolved in hot acetone. The crystalline product which separated from the cold solution melted at 239-240°C.

Step 2: The mono-substitution product obtained in Step 1 could be employed as the free base or hydrochloride for additional reaction with other secondary amines and aldehydes or ketones as previously described. Again, reaction techniques and conditions follow the same as outlined. The products obtained have the nitrogen-methano groupings on the two alpha carbon atoms joined to the cyclic carbonyl, e.g., 2-(dibenzylaminomethyl)-6-(dimethylaminomethyl)-cyclohexanone and its dihydrochloride.

Example 7

2,5-Dimorpholinomethylcyclopentanone

This compound was prepared according to the method described in R.H. Harradance et al, Royal Soc. New S. Wales Proc. Vol. 72, p. 245 (1938) by refluxing together 21 g. cyclopentanone, 31 g morpholine hydrochloride, 9 g. paraformaldehyde and 20 ml absolute alcohol for several hours, which led to the gradual separation of a white crystalline substance. After cooling, this was collected and boiled with alcohol, in which it was very sparingly soluble. Recrystallized from aqueous alcohol, it formed colorless needles m.p. 195°C.

The following Examples 8 to 12 describe intermediates which can be em-

ployed to form mixed amino keto bases and acid salts by further reaction at the remaining active hydrogen with different secondary amines and formaldehyde (or its polymer) to yield compounds of formula I:

Example 8

Piperidino-methyl-cyclopentanone-2 Hydrochloride

This is prepared according to C. Mannich et al., Archiv. Pharm. 276, 578 (1938) by heating a mixture of 6 g. piperidine hydrochloride, 3.7 g. 40% aq. formaldehyde solution and 8.4 g. cyclopentanone on a water bath under a reflux condenser. After 25 minutes the reaction was finished. The excess cyclopentanone was removed by ether extraction and the reaction product evaporated to dryness in vacuo; yield 90% on recrystallization from isopropyl alcohol, colorless needles m.p. 145°C with decomposition.

Example 9

2-Morpholinomethylpentan-3-one Hydrochloride

Following the directions in the publication by R.H. Harradence et al, Roy. Soc. New S. Wales Proc. 73, 20 (1939), 22 g. Diethyl ketone, 7.5 g. paraformaldehyde, and 31 g. morpholine hydrochloride were refluxed on the water bath for 5 hrs. The hydrochloride was then precipitated with ether and filtered off. Recrystallized from acetone, it formed colorless needles, m.p. 131°C. The free base was isolated by neutralization in the usual manner, b.p. 95-100°/2 mm.

Example 10

1-Morpholinobutan-3-One

In accordance with the description in R.H. Harradence et al., Roy. Soc. New S. Wales Proc., vol. 72, 237 (1938), 60 cc acetone, 42 g. morpholine hydrochloride, 16.8 g. paraformaldehyde and 10 ml. absolute alcohol were heated together under reflux for six hours. The solid gradually disappeared and eventually a homogeneous solution was obtained. On shaking, this set to a solid crystalline mass while still hot. The mass was cooled, and the crystals freed from excess acetone on a porous plate. Recrystallized from absolute alcohol, 1-morpholinobutan-3-one hydrochloride was obtained, colorless crystals, m.p. 149°C.

The free base was prepared from the crude hydrochloride after removal of excess acetone by making the solution strongly alkaline and extracting with ether. After drying and removal of the solvent, the oily base was distilled in vacuo. It came over as a colorless oil, 38 g., b.p.116°C/20 mm.

Example 11

2-Morpholinomethyl cyclohexanone

Following the teaching on pp. 240-1 of the last cited publication, 49g. cyclohexanone, 13 g. morpholine hydrochloride and 9 g. 40% aq. formaldehyde solution were gently warmed and vigorously shaken together. After about 10 minutes, some heat was evolved and eventually a homogeneous liquid was obtained. This was heated at 100°C for a further 5 minutes, then allowed to cool.

Example 12

1-N-Piperidino-2-methylbutanone-3

Pursuant to the description in C. Mannich et al, Archiv. Pharmazia, 265, p. 594 (1927), 6.1 g. piperidine hydrochloride, 42 g. methyl ethyl ketone and 2.1 g. paraformaldehyde were heated until, with the gradual addition of 5 cc absolute alcohol, complete solution was obtained. The excess methyl ethyl ketone was removed by distillation. On cooling, the hydrochloride of the basic ketone crystallized. It was recrystallized from a mixture of acetone and alcohol, m.p. 150°C. The free base obtained from this salt had a b.p. 103-104°/14 mm.

This compound could be further substituted at the active hydrogen by reaction with secondary amine and formaldehyde or its polymer to yield compounds corresponding to formula I.

A further example of the formation of mixed amino keto bases and their salts is the following:

Example 13

2-Dimethylaminomethyl)-6-(Piperidinomethyl)-Cyclohexanone Dihydrochloride

Following the instructions in F.F. Blicke et al, J. Org. Chem., 24, p. 1074 (1959), a mixture of 19.2 g. of 2-(dimethylaminomethyl) cyclohexanone hydrochloride, 3.0 g. paraformaldehyde, 12.2 g. of piperidine hydrochloride and 40 ml. of acetic acid were maintained at 95°C for 2.5 hours, using agitation to obtain, finally, a clear solution. The solvent was removed on a steam bath under reduced pressure and the product, 15.5 g., was recrystallized from methanol, ether, m.p. 168-169°C.

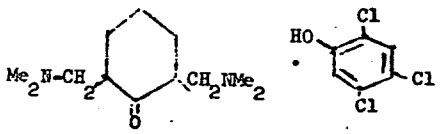
Additional examples of identical and mixed amino keto bases and their acid salts obtained from the appropriate secondary amine, "parent" ketone, and aldehyde, preferably formaldehyde, or ketone devoid of active hydrogens, as will be evident from the foregoing, are the following: 2-(dimethylamino-

omethyl)-6(morpholinomethyl)-cyclohexanone, 2(dimethylaminomethyl)-6-(thiazolinomethyl)-cyclohexanone, 2-(dimethylaminomethyl-6-(pyrrolidinomethyl)-cyclohexanone, 2-(diethylaminomethyl)-6-(piperazinomethyl)-cyclohexanone, 2,4-bis-(dimethylaminomethyl)-cyclobutanone, 2-Morpholinomethyl-5-dimethylaminomethyl-cyclopentanone, 2,8-bis-(diethylaminomethyl)-cyclooctanone, and 2,4-bis-(dimethylaminomethyl)-2,4-dimethyl-pentanone-3, and their mono- and di-hydrochlorides, hydrobromides and acetates, sulfates, and other acid salts and their quaternary salts obtainable by conventional procedures. Among the known condensates that are antimicrobially active may be mentioned also 2,7-bis-(dimethylaminomethyl) cycloheptanone, 2,5-bis-(piperidinomethyl) cyclopentanone, 2,6-bis-piperidino-methyl) cyclohexanone, 2,6-bis-(benzylmethyl-aminomethyl)-cyclohexanone, and 2,6-bis-(dibenzylaminomethyl)-cyclohexanone and their salts.

The following examples illustrate the formation of salts with substituted phenols:

Example 14

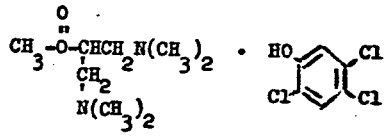
2,6-Bis-(Dimethylaminomethyl)-Cyclohexanone • Mono-2,4,5-Trichlorphenate



100 g. of 2,4,5-trichlorphenol were dissolved in 300 g. of ethylene glycol and, with good agitation, 100 g. of the diamine were added to give a 40% solution of the salt.

Example 15

1-Dimethylamino-2-Dimethylaminomethyl-Butanone-3 Mono-2,4,5-Trichlorphenate Morohydrochloride



20 g. of a solution containing 16.1 g. (0.068 M equivalent) of the 1-dimethylaminomethyl-2-dimethylaminomethyl butanone-3 dihydrochloride were added to 63.2 g. of ethylene glycol. Neutralization was effected with 2.7 g. of NaOH and 14.1 g. of 2,4,5-trichlorphenol were added.

In the following examples there will be employed the abbreviations BC for 1-(dimethylamin)-2-(dimethylaminomethyl) butanone-3 and BDMC for 2,6-bis-(dimethylaminomethyl)-cyclohexanone.

Example 16

BC-o-Phenyl Phenate Monohydrochloride

20 g. of a solution containing 16.1 g. (0.068 M equivalent) of BC.2HCl were added to 65.5 g. of ethylene glycol. Neutralization was effected with 2.7 g. of NaOH. 11.8 g. of o-Phenyl phenol were then added and the mixture stirred until solution took place. After filtration, to remove small amounts of insolubles, a clear, stable solution was obtained.

Example 17

BDMC.2,4,6-Trichlorphenate

21.2 g. (0.1 M) of BDMC were dissolved in 60 g. of "Solox" (ethanol). To this solution were added 19.7 g. (0.1 M) of 2,4,6-trichlorphenol. Agitation gave a clear solution containing about 40% BDMC.mono-2,4,6-trichlorphenate.

Example 18

BDMC·di-(2,4,6-Trichlorphenate)

39.4 g. (0.2 M) of 2,4,6-trichlorphenol were dissolved in 40 g. of ethanol. Then, with agitation, there were added 21.2 g. (0.1 M) of BDMC. All dissolved rapidly to give a clear solution containing approximately 60% of BDMC-di-2,4,6-trichlorphenate.

Example 19

BDMC · di-(2,4,5-Trichlorphenate)

10.6 g. of BDMC were dissolved in 38 cc of ethanol containing 9.9 g. of 2,4,5-trichlorphenol to give a 60% solution of BDMC· di-(2,4,5-trichlorphenate).

Example 20

BDMC-o-Phenyl Phenate

To a solution containing 17.0 g. (0.1 M) of o-phenyl phenate in 61.8 g. of ethanol, there were added 21.2 g. (0.1 M) of BDMC. Agitation yielded a clear solution containing approximately 38% BDMC mono-o-phenyl phenate.

Example 21

BDMC-di(o-Phenyl Phenate)

34.0 g. 0.2 M) of o-phenyl phenol were dissolved in 44.8 g. of ethanol. Then 21.2 g. (0.1 M) of BDMC were added to give a solution containing approximately 65% BDMC di-(o-phenyl phenate).

The compounds, in which one or more of X_1 to X_4 in formulas I to III hereinabove are hydrogen, are not prepared by the use of primary amines, as

these would lead to side reactions by reason of the additional active hydrogens. They are preferably prepared by the use of secondary amines whose substituents can be removed in known manner after the condensation reaction with the carbonyl-containing compounds. Thus by the use of dibenzylamine, followed by debenzylation according to known procedures, as by treatment with hydrogen in the presence of a palladium catalyst in hydrochloric acid solution, compounds can be obtained in which all of X_1 to X_4 are hydrogen; while by the use of ethyl-benzylamine, followed by debenzylation, one hydrogen will be attached to each nitrogen atom. By the use of a two-step process, using first, for example, dibenzylamine, and methylbenzylamine in the second step, followed by debenzylation, products of formulas I to III in which three of the groups X_1 to X_4 are hydrogen; while if, for example, dimethylamine or diethylamine is employed in the second step, two hydrogens will be attached to the same nitrogen. These di-primary amine, di-secondary amine, and mixed primary-secondary, primary-tertiary and secondary-tertiary amine compounds so obtained can be quaternized or converted into salts with acids and phenols as described above in connection with the tertiary amine and products.

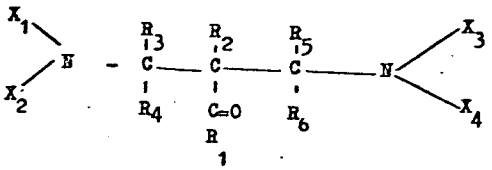
Although formaldehyde and its polymers are the preferred aldehyde compounds for linking the "parent" ketone with the amine, other aldehydes like α-plurally substituted acetaldehyde and butyraldehyde, benzaldehyde, and the like, can be substituted for the formaldehyde. Similarly, with higher aliphatic aldehydes utilized in place of the formaldehyde, those will yield the simpler condensates conforming to formulas I, II and III which are sufficiently substituted at the α-carbon to eliminate active hydrogen. The formaldehyde can be replaced also by ketones free of reactive hydrogens, i.e., by hexa-substituted acetones, such as hexamethyl and hexaethyl acetone, benzyl-pentaethyl acetone, cyclopentyl-pentaethyl acetone, cyclohexyl-pentamethyl acetone, s-dibenzyl tetramethyl acetone, and the like. The substituents can include, in addition to alkyl groups, also chlorine, bromine, NO_2, hydroxyl, and other non-interfering groups.

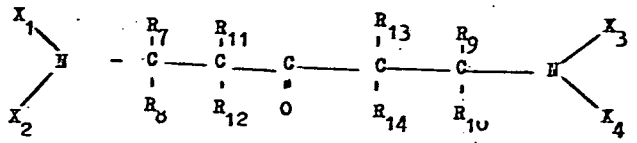
The use of the above described condensates in paints has reference particularly to water-base paints, like polyvinyl acetate and other paints wherein the filming or suspending agents are organic in nature and are attacked by bacteria. One part, for example, 2,6-bis-(dimethylaminomethyl)-cyclohexanone in 4000 parts of the paint is sufficient to protect the paint

against such attack. The above-described products can also be used for protecting resinous suspensions of floor waxes and polishes and similar products.

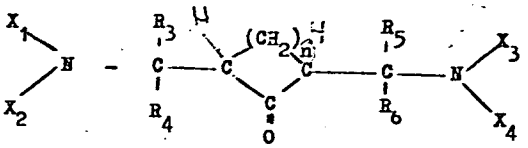
The claims defining the invention are as follows:

1. The method of protecting microbe-susceptible materials against the deterioration action of microbes, which comprise applying to such materials a small but effective proportion of a member of the group consisting oc compounds of the formulae:





and



and their salts, wherein:

R_1 to R_{14} are each a member of the group consisting of hydrogen, alkyl having up to 18 carbons, alkenyl having up to 18 carbons, aralkyl and cycloalkyl having up to 9 carbons, except that at least one of R_{11} to R_{14} is not hydrogen; while R_2 can also be lower hydroxyalkyl;

X_1 to X_4 are each a member of the group consisting of hydrogen, alkyl having up to 18 carbons, alkenyl having up to 18 carbons, hydroxyalkyl, cycloalkyl having up to 9 carbons, mononuclear aryl and aralkyl having up to 9 carbons, while at least one of the pairs of X_1 and X_2, and X_3 and X_4, may be part of a saturated heterocyclic ring containing the indicated N and together with carbon also a further hetero atom of the group consisting of oxygen, nitrogen and sulfur; and n is an integer from 1 to 6 inclusive. (8th January, 1962)

2. The method of protecting microbe-susceptible materials against against the deteriorating action of microbes according to claim 1, characterized by the fact that there is applied to such materials such but effective portion of 2,2'-bis-(dimethylaminomethyl)-cyclohexanone, 2,2'-bis-(diethylaminomethyl)-cyclohexanone, 2,2'-bis-(aminomethyl)-cyclohexanone, 1,1-bis-(di

ethylaminomethyl)-acetone, or 1,1-bis-(dimethylaminomethyl)-acetone. (8th January, 1962)

3. The method of protecting microbe-susceptible materials against the deteriorating action of microbes, characterized by the fact that there is applied to such materials a substituted phenol salt of the defined base, according to claim 1. (8th January, 1962)

4. The method of protecting microbe-susceptible materials against the deteriorating action of microbes, according to claim 1, characterized by the fact that there is applied to such materials a 2,4,5- or a 2,4,6-trichlorophenol salt of 2,2-bis-(dimethylaminomethyl)-cyclohexanone, 1,1-bis-(diethylaminomethyl)-acetone, or 1,1-bis-(dimethylaminomethyl)-acetone. (8th January, 1962)

5. The method of protecting microbe-susceptible materials against the deteriorating action of microbes, characterized by the fact that there is applied to an industrial hydrocarbon oil-water emulsion a small but effective proportion of a compound as defined in claim 1. (8th January, 1962)

6. The method of claim 5, characterized in that the compound as defined in claim 1 is 2,2'-bis-(dimethylaminomethyl)-cyclohexanone. (8th January, 1962)

7. The method of protecting microbe-susceptible materials against the deteriorating action of microbes, characterized by the fact that there is applied to waters employed in pulp and paper manufacture for suspending the pulp, a small but effective proportion of a compound as defined in claim 1. (8th January, 1962)



8. The method of claim 7, characterized in that the compound as defined in claim 1 is 2,2'-bis-(dimethylaminomethyl)-cyclohexanone. (8th January, 1962)

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9. A microbiologically controlled product comprising a microbe-susceptible material containing a small but effective proportion of a compound as defined in claim 1. (8th January, 1962)

10. A microbiologically controlled paper pulp suspension or cutting oil composition containing a small but effective proportion of a compound as defined in claim 1. (8th January, 1962)

11. A microbiologically controlled paper pulp suspension or cutting oil composition according to claim 10, characterized in that the compound as defined in claim 1 is 2,2'-bis-(dimethylaminomethyl)-cyclohexanone. (8th January, 1962)

12. A microbiologically controlled product containing a small but effective proportion of a substituted phenol salt of a compound as defined in claim 1. (8th January, 1962)

13. A microbiologically controlled product according to claim 12, characterized in that the substituted phenol salt of the compound as defined in claim 1 is a chlorophenol salt of such compound. (8th January, 1962)

14. A microbiologically controlled product according to claim 13, characterized in that the chlorophenol salt of the compound as defined in claim 1 is a 2,4,5-trichlorophenol salt of such compound. (8th January, 1962)

15. A microbiologically controlled product according to claim 14, characterized in that the 2,4,5-trichlorophenol salt of the compound as defined in claim 1 is the 2,4,5-trichlorophenol salt of 2,2'-bis-(dimethylaminomethyl)-cyclohexanone. (8th January, 1962)

16. A 2,4,5-trichlorophenol salt of 2,6-bis-(dimethylaminomethyl)-cyclohexanone. (8th January, 1962)

17. A 2,4,6-trichlorophenol salt of a compound as defined in claim 1. (8th January, 1962)

18. The o-phenyl phenol salt of a compound as defined in claim 1. (8th January, 1962)

19. Process for the manufacture of antimicrobial agents, substantially as described with references to the preceding examples. (8th January, 1962)

20. Microbe-susceptible materials when protected against microbial action by any of the processes or with any of the substances claimed in any of the preceding claims. (8th January, 1962)

DATED this SIXTH day of JANUARY, 1967

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